P-C-437A
March 11, 1970

SUPERSEDING
Fed. Spec. P-C-437
October 30, 1961

FEDERAL SPECIFICATION

CLEANING COMPOUND, HIGH PRESSURE (STEAM) CLEANER

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE

1.1 This specification covers alkaline steam cleaning compound for use in steam cleaning machines for cleaning various ferrous and nonferrous surfaces.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

Federal Specifications:

L-P-378	-	Plastic Sheet and Strip (Polyolefin).
O-S-604	-	Sodium Metasilicate, Technical.
QQ-A-250/1	-	Aluminum 1100, Plate and Sheet.
QQ-A-250/4	-	Aluminum Alloy 2024, Plate and Sheet.
PPP-D-705	-	Drum: Metal Shipping, Steel (Over 12 and Under 55 Gallon).
PPP-D-723	-	Drums, Fiber.
PPP-D-729	-	Drums: Metal, 55 Gallon (For Shipment of Noncorrosive Material).
PPP-D-732	-	Drums, Metal, 55 Gallon Reconditioned, (For Shipment of Noncorrosive Materials).
PPP-P-704	-	Pails, Metal: (Shipping, Steel, 1 through 12 Gallon).

Federal Standards:

Fed. Std. No. 123	-	Marking for Domestic Shipment (Civilian Agencies).
Fed. Test Method Std. No. 536	-	Soap and Soap Products (Including Synthetic Detergents); Sampling and Testing.

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(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, Washington.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specifications:

MIL-P-116 - Preservation, Methods of.
MIL-L-15016 - Lubricating Oil, General Purpose.
MIL-D-16791 - Detergents, General Purpose (Liquid Nonionic).

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM) Standards:

D 502 - Particle Size of Soaps and Other Detergents.
E 11 - Sieves for Testing Purposes (Wire-Cloth and Perforated-Plate Sieves).

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

(Technical societies and technical association specification and standards are generally available for reference from libraries. They are also widely distributed among technical groups and using Federal agencies.)

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3. REQUIREMENTS

3.1 Qualification. The cleaning compounds furnished under this specification shall be products which are qualified for listing on the applicable Qualified Products List at the time set for opening of bids (see 4.3 and 6.3). Any change in the formulation of a qualified product will necessitate its requalification. The material supplied under contract shall be identical, within manufacturing tolerances, to the product receiving qualification.

3.2 Material. The ingredient materials used in the manufacture of the steam cleaning compound covered by this specification shall be of high quality, intimately assembled and processed so as to produce a nonhygroscopic, nonsegregating, granular, free-flowing mixture which will show no evidence of caking in the as-received condition or during storage. When used in a steam cleaning machine of the high pressure-continuous tubular coil vapor generator type, the compound shall not impede nor clog the flow of solution by precipitation and building up of solids in the jets, coils, or orifices. The compound shall not adversely affect the user (skin burn or sneezing).

3.2.1 Condition as-received. The steam cleaning compound in the as-received condition shall show no evidence of caking (for example, lumping or agglomerating) when tested as specified in 4.4.15.

3.2.2 Ingredient materials. The ingredient materials used in the manufacture of the steam cleaning compound covered by this specification shall be granular, not powdered. For this purpose, a powdered material is considered to be one whose particles have a spherical rounded shape, while a granular material is characterized by particles with a nonspherical, irregular shape with nonrounded surfaces. Each nonliquid ingredient shall be examined under 10 power magnification for compliance with this requirement.

3.2.3 Synthetic detergents. All anionic and nonionic detergents used in the manufacture of this cleaning compound shall be at least 80 percent biodegradable. An affidavit to this effect shall be furnished by the manufacturer with each lot of cleaning compound submitted for inspection and with the qualification sample.

3.3 Composition. The steam cleaning compound is not required to conform to definite chemical composition requirements. The manufacturer is given latitude in the selection of raw materials and processes of manufacturer, provided the product meets all applicable requirements of this specification. The compound shall be free from fatty acid, rosin, soaps, starch, abrasives, gritty material, inert fillers, carbonates, bicarbonates, and free sodium hydroxide.

3.3.1 Carbonates or bicarbonates. The compound shall show no effervescence, when tested as specified in 4.4.1.

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3.3.2 Fatty acid, soap, rosin, and starch content. The compound shall give no evidence of the presence of fatty acid, rosin, soap, or starch when tested in accordance with 4.4.2.

3.3.3 Free sodium hydroxide. When tested as specified in 4.4.3, not more than 0.2 ml of 0.1 normal HCl shall be required to discharge any pink coloration.

3.4 Performance requirements. The steam cleaning compound shall be equal to or superior in effectiveness to the comparison formula in 4.4.4.1.3 when tested as specified in 4.4.4.1 and 4.4.4.2. The compound covered by this specification shall be tested simultaneously with the standard comparison steam cleaning compound specified in 4.4.4.1.3.

3.4.1 Water softening or stability. When tested as specified in 4.4.4.1 at a temperature of 260 - 265°F., a one-quarter percent solution of steam cleaning compound in water of 20 grain hardness shall give no evidence of precipitation or curd formation; nor shall there be an opalescence present greater than that produced in a solution of the standard comparison compound specified in 4.4.4.1.3 when tested simultaneously with the test steam cleaning compound covered by this specification. The opalescence shall be not greater than that given by the statistical upper limit set forth in 4.4.4.1.5.

3.4.2 Cleaning efficiency. The cleaning compound shall be equal to or superior to the standard comparison steam cleaning compound in ability to remove a standard soil when tested as specified in 4.4.4.2. The residual soil of any test compound determination shall be not greater than the statistical upper limit as set forth in 4.4.4.1.5.

3.5 Solubility. The insoluble matter in a 10 percent distilled water solution of the compound shall not exceed 0.25 percent when tested as specified in 4.4.5.

3.6 pH value. The pH of a one-half percent solution of the cleaning compound in distilled water shall be not less than 10.5 nor more than 11.4, when tested as specified in 4.4.6.

3.7 Corrosion. Boiling solutions of the cleaning compound in distilled water shall cause no loss in weight of aluminum or aluminum alloy test specimens in excess of that indicated in table I when tested as specified in 4.4.7. There shall be no visible staining, discoloration, etching, or pitting of the test specimens.

TABLE I. Loss in weight permitted in corrosion test

Percent solution	Fed. Spec. QQ-A-250/1 Al-1100	Fed. Spec. QQ-A-250/4 Al-2024
0.25	1.2 mg	0.8 mg
1.2	0.8 mg	0.5 mg

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3.8 Surface tension. The surface tension of a one-quarter percent solution of the compound in distilled water shall be not more than 42 dynes per centimeter when determined as specified in 4.4.8.

3.9 Dust forming properties. The dust forming properties shall be such that, when tested as specified in 4.4.9, the dust shall settle within a period of five seconds.

3.10 Penetration. When tested as specified in 4.4.10, the average penetration shall be not less than 30.0.

3.11 Segregation. When tested as specified in 4.4.11, the percent of any ingredient of the compound (such as silicates or total phosphates) taken from different portions of a container shall not differ by more than 2 percent from the average percentage of that ingredient in the compound. The compound shall be subjected to the test specified in 4.4.11 at the discretion of the inspection laboratory when segregation appears to have occurred.

3.12 Fineness. The particle size of the cleaning compound shall be such that 100 percent shall pass a No. 6 sieve and not more than 3.0 percent shall pass a No. 100 sieve when tested as specified in 4.4.12.

3.13 Rinsing. Solutions of the compound shall leave no white film on aluminum alloy when tested as specified in 4.4.13.

3.14 Caking in storage. When tested as specified in 4.4.14, the compound shall remain free flowing and of such a texture and granulation that 100 percent shall pass a No. 6 U. S. Standard sieve.

3.15 Workmanship. The steam cleaning compound shall be manufactured in accordance with the best commercial practice to produce a high quality, stable, nonhygroscopic, nonsegregating, granular, free-flowing mixture.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling and inspection.

4.2.1 Lot. For purposes of sampling, a lot shall consist of material mixed or prepared at one time. Material shall be identified by order of production or batch number.

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4.2.2 Sampling procedure. From each inspection lot three separate 1-pound samples shall be taken. In case the material is produced by a batch process and the inspection lot contains more than 2 batches, the three samples shall be taken from different batches. Samples shall be obtained in a manner calculated to disclose any nonuniformity of the material within the batch. Where material is produced by a continuous-run process, the three samples shall be taken so as to represent, respectively, the first part, the middle part, and the last part of the run which produced the inspection lot.

4.2.3 Sampling for inspection of filled containers. A random sample of filled containers shall be taken from each lot in accordance with MIL-STD-105 at inspection level 1, and acceptable quality level equal to 2.5 percent defective, to verify compliance with all stipulations of this specification regarding fill, closure, marking and other requirements not involving tests.

4.2.4 Inspection of filled containers. Each sample filled container, selected in accordance with 4.2.3, shall be examined for defects of the container and the closure, for evidence of leakage, and for unsatisfactory markings; each sample filled container shall also be weighed to determine the amount of the contents. Any container in the sample havine one or more defects, or under required fill, shall be rejected; and if the number of defective containers in any sample exceeds the acceptance number for the appropriate sampling plan of MIL-STD-105, the lot represented by the sample shall be rejected.

4.3 Classification of tests. Testing under this specification shall be for the following:

(a) Qualification.

(b) Acceptance of individual lots.

4.3.1 Qualification tests (see 6.3). Qualification tests shall consist of tests for all the requirements of this specification. For qualification testing a 10-pound sample is required plus a 125-pound sample in a metal drum as required for the caking in storage test. Separate two-pound samples of each nonliquid ingredient shall also be submitted for qualification testing (see 3.2.2). Interim approval will be granted a product passing all requirements of the specification except 3.14, subject to recision if 3.14 is failed.

4.3.2 Acceptance tests. Inspection tests for acceptance of individual lots shall normally consist of tests for water softening, soap and rosin contents, pH, carbonates, and 1100 aluminum - 1/4 percent solution corrosion. If any sample is found to be not in compliance with this specification, the entire lot shall be rejected.

4.4 Test methods.

4.4.1 Carbonates or bicarbonates. Approximately 6 normal hydrochloric acid shall be added drop-wise to a five percent (5 grams of compound in 100 ml of solution) distilled water solution of the compound and examined for effervescence.

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4.4.2 Fatty acid, soap, rosin, and starch content.

4.4.2.1 Fatty acid, soap, rosin. Place 10 grams of the compound in a 100-ml tall-form beaker. Add 30 ml absolute ethyl alcohol. Place on a steam plate and stir for 5 minutes. Filter through dry, double, No. 41 Whatman papers. Evaporate filtrate to dryness. Dissolve residue in 20 ml distilled water. To half the solution add 5 ml of 1.0 N acetic acid. To the other half, add 5 ml of 1.0 N calcium chloride solution. Formation of the precipitate in each solution indicates presence of soap, rosin or fatty acid.

4.4.2.2 Starch. Starch content shall be determined by method 6401, note 1, Fed. Test Method Std. No. 536.

4.4.3 Free sodium hydroxide. Place 10 grams of compound in a 100-ml tall-form beaker. Add 30 ml petroleum ether (B.P. 35-60° C), and stir 3 minutes with a glass rod. Decant off petroleum ether. Place on steam bath for five minutes to drive off hazardous gases, then dry in oven for fifteen minutes at 105° F. Add 30 ml absolute ethyl alcohol and stir 2 minutes. Decant off alcohol through dry, triple, No. 41 Whatman filters. Transfer 15 ml of the filtrate to a 125 ml flask and add several drops phenolphthalein indicator solution. If pink coloration results, titrate with 0.1 normal hydrochloric acid. Compare titration with requirements of 3.3.3.

(a) Water softening or stability tests.

(b) Cleaning efficiency.

4.4.4.1 Water softening or stability tests.

4.4.4.1.1 Preparation of hard water stock solution. A 40-grain per gallon hard water solution is prepared by dissolving 0.8090 grams of ACS grade calcium acetate, $Ca(C_2H_3O_2)2 \cdot H_2O$, and 0.5546 grams of ACS grade magnesium sulphate, $MgSO_4 \cdot 7H_2O$, per liter of boiled distilled water.

4.4.4.1.2 Preparation of cleaning compound solution. Weigh 10 grams of the cleaning compound into a 2-liter volumetric flask. Add 500 milliliters of hot water 190°F) and agitate to dissolve the cleaning compound. Cool to room temperature and dilute to volume with distilled water.

4.4.4.1.3 Preparation of standard comparison cleaning compound solution. Using a cleaning compound having a composition conforming to table II, prepare a standard cleaning compound solution as specified in 4.4.4.1.2 to be used as the comparison standard.

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TABLE II. Composition of standard comparison compound

Component	Percent by weight
Sodium metasilicate, pentahydrate, Fed. Spec. O-S-604	35.0
Primary sodium phosphate, ACS grade $NaH_2PO_4 \cdot H_2O$	10.5
Sodium tripolyphosphate, technical grade $Na_5P_3O_{10}$	52.5
Nonionic surface-active agent, MIL-D-16791, type 1	2.0

4.4.4.1.4 Water softening test. To a test tube (200 mm long by 25 mm outside diameter) containing 20 ml of the cleaning compound solution (4.4.4.1.2), add 20 ml of hard water stock solution (4.4.4.1.1), and mix well. Immerse in the solution, a test panel of 1020, 20-gage, cold-rolled steel, 3 inches by 1/2 inch in size, which has been polished with No. 1A3 emery cloth and cleaned with acetone. Place the test tube in a laboratory steam sterilizer at 260° - 265° F (steam pressure of 21-24 lb. per sq. in.) for 1 hour. Remove from the sterilizer. Cool to room temperature, and examine for precipitation. Use an electrophotometer to measure opalescence in terms of optical density. Tests shall be run in quadruplicate. The same test shall be run in quadruplicate using the standard comparison compound (4.4.4.1.3).

4.4.4.1.5 Statistical upper limit. The statistical upper limit at the 0.05 level in the water softening and cleaning tests of 4.4.4.1.4 and 4.4.4.2.4 may be determined with the following formula:

$$X + 0.72W$$

where: X = average of 4 determinations with the comparison compound.

W = difference between the maximum and minimum values in the four determinations with the comparison compound.

4.4.4.2 Cleaning efficiency test.

4.4.4.2.1 Preparation of test panels. Test panels of 1020, 20-gage, cold-rolled steel shall be prepared 2-1/2 by 2-1/2 inches in size. Each test panel shall have a hole 1/4 inch in diameter placed 1/8 inch from one corner. Sharp edges shall be smoothed with No. 1A3 emery cloth. The panels shall then be degreased in ACS grade acetone. Both faces of each panel shall be polished with the same abrasive, stroking in one direction only. Clean in a hot alkaline solution until free from water-break. Rinse in water, then dip in absolute ethyl alcohol, and wipe dry with paper toweling.

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4.4.4.2.2 Standard soil. The soiling material shall be Military symbol oil 3050 conforming to MIL-L-15016. The cleaned panels, prepared as described in 4.4.4.2.1, are suspended on a "S" hook and dipped one at a time into a 400-ml beaker of the oil maintained at a temperature of 25 ± 1°C. They are then removed and allowed to drain at a temperature of 25 ± 1°C for 30 minutes. The remaining drop of oil in the lower corner of each panel is then removed with absorbent cotton. The amount of mineral oil adhering per panel shall be approximately 0.16 grams.

4.4.4.2.3 Preparation of cleaning solutions. In a 2-liter beaker, place 1,600 ml of 1/2 percent distilled water solution of the test compound (see 4.4.4.1.2). Bring solutions to a boil and keep at this temperature throughout the test. Maintain solution level throughout test by addition of distilled water as required.

4.4.4.2.4 Cleaning procedure. Immerse the soiled test panels, prepared as specified in 4.4.4.2.2, in the cleaning solution and in the comparison solution by means of an iron or copper hook, one end of which passes through the 1/4-inch hole and the other end over a glass rod placed across the top of the 2-liter beaker. At the end of 2-1/2 minute and 5 minute intervals, respectively, the test panel is moved forward and backward, 3 times in each direction, and then up and down, 3 times in each direction; total agitation shall require not more than 6 seconds. At the end of 5 minutes immersion, the panel is removed from the cleaning solution and given 2 six second rinses in distilled water (no agitation); there shall be a four second drain between rinses. The rinsing solutions shall consist of 2 one liter beakers, each containing 800 ml of distilled water at 25 ± 1°C. The panel is then dried at 50°C for 20 minutes, cooled and weighed. The panel is thoroughly washed with acetone, rinsed in absolute ethyl alcohol, dried with paper toweling and reweighed. The difference in weight is the amount of residual soil. Tests shall be run in quadruplicate.

4.4.5 Solubility. Weigh 10 grams of the compound and add to 100 ml of distilled water at 25 ± 1°C in a 300-ml, tall-form beaker. Stir until compound is thoroughly dissolved. Let stand 10 minutes. Filter through a tared Gooch crucible with an asbestos mat. Wash with distilled water until washings give no alkaline reaction to phenolphthalein. Dry at 105°C for 3 hours, cool and weigh. Gain in weight of crucible and contents is insoluble matter.

4.4.6 pH value.

4.4.6.1 Apparatus. The test shall be made with a pH electrometer having a sensitivity and readability of at least 0.05 pH. A sealed type, alkali-resistant glass electrode shall be used with a calomel reference electrode. The meter shall be standardized against a pH 10 buffer solution immediately before making a test. The instrument to be used shall be the Beckman pH Meter Laboratory Model G, or equal, and the electrode shall be the Beckman "Type E" high pH glass electrode or equal.

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4.4.6.2 Procedure. Prepare 100 ml of a 1/2 percent (see 4.4.4.1.2) distilled water solution of the compound. Determine pH at 25 ± 1°C. No correction for sodium ion concentration shall be made.

4.4.7 Corrosion test.

4.4.7.1 Preparation of test panels. Test panels of 1100 aluminum shall be cut 3 inches by 3/4 inch in size from approximately 0.034 inch thick sheet aluminum conforming to QQ-A-250/1, H14 or H24 temper. Test panels of 2024-T3 aluminum shall be cut 3 inches by 3/4 inch in size from approximately 0.025 inch thick sheet conforming to QQ-A-250/4. Sharp edges shall be smoothed with No. 4/0-150 alundum (Al_2O_3) polishing paper. Polish both faces with the same abrasive, stroking in one direction only. The 1100 aluminum panels are not to be reused in this test. The 1100 aluminum panels should be lightly polished to avoid smearing of the surface, with approximately 18 milligrams of metal removed in polishing.

4.4.7.2 Cleaning of panels. Panels shall be cleaned with ACS grade acetone using a swab of absorbent cotton. They shall then be wiped with paper toweling, dipped in absolute ethyl alcohol, and again wiped with paper toweling.

4.4.7.3 Preparation of test solutions. Weigh 2.5 gram of the cleaning compound into a 1 liter volumetric flask. Add 500 ml of hot water (190°F) and agitate to dissolve the cleaning compound. Cool to room temperature and dilute to volume with distilled water. Prepare another solution in the same manner except using 12 grams of cleaning compound.

4.4.7.4 Corrosion test procedure. Each cleaned test panel shall be weighed to 0.0001 gram and immersed completely in 200 ml of the boiling solution of the compound in a 500-ml tall-form beaker. The solution is kept at a boil throughout the test, the solution level being maintained by additions of distilled water as required. After 60 minutes, the test panel shall be removed from the boiling solution, rinsed under flowing tap water (cold), rinsed in distilled water (25 ± 1°C), dipped in absolute ethyl alcohol, wiped dry with paper toweling, and reweighed. The presence of corrosion products shall be noted. Tests shall be run in duplicate on each aluminum material in each solution.

4.4.8 Surface tension. Using 100 ml of the 1/4 percent distilled water solution of the compound prepared as in 4.4.7.3, determine the surface tension at 25 ± 1°C using a DuNouy tensiometer or interfacial tensiometer. Harkins-Jordan correction factors shall be applied.

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4.4.9 Dust forming. The inside walls of a 250-ml glass-stoppered graduate shall be rendered completely free of any grease and moisture. A 25-gram sample of the cleaner shall then be placed on the bottom of the clean graduate. The graduate shall then be stoppered, inverted and immediately returned to its original position. After any suspended dust has been allowed to settle for 5 seconds, the stopper of the graduate shall be removed and a moistened piece of red litmus paper suspended in the graduate without touching the walls so that the lower end of the litmus paper strip coincides with the 210 ml mark at the upper end of the graduate. After 60 seconds, the litmus paper shall be removed and examined for any color change. The test shall be run in a constant humidity room, at a relative humidity of 50-55 percent and a temperature of 25 ± 1°C.

4.4.10 Penetration. Place 100 grams of the compound in a flat bottom, straight side, Pyrex glass crystallizing dish, 40 mm high by 80 mm diameter. Place sufficient lead shot into a 250-ml low-form beaker so that the combined weight of the beaker and shot equals 800 grams. Place the weighted 250-ml beaker on top of the 100 grams of compound. Place the assembly in a desiccator containing a saturated solution of potassium sulphate (K_2SO_4). Place the desiccator in an oven maintained at 43 - 46°C for 24 hours. Remove the assembly from oven and desiccator; remove weighted 250-ml beaker from compound, and allow crystallizing dish containing compound to cool at room temperature for 2 hours. Determine penetration at three points on the surface, each at least 12 mm from the side of the crystallizing dish, using the penetrometer and grease-penetrometer cone prescribed in ASTM D217 with a 5-second hold, 250 grams total weight of the cone and attachments and the temperature of the compound at 25 ± 1°C. Run test in duplicate.

4.4.11 Segregation. Samples taken from different portions of the container shall be analyzed for one constituent of the compound (such as total phosphates or silicates) in accordance with the applicable method of Fed. Test Method Std. No. 536.

4.4.12 Fineness. A No. 6 and a No. 100 sieve shall be used in a Ro-Tap (or equal) sieve shaking apparatus. The sieves shall be 8-inch diameter U. S. Standard wire cloth sieves conforming to ASTM E11. Transfer a 100 ± 0.1 gram sample, without previous drying, to the No. 6 sieve. Allow the shaking apparatus to run 10 minutes. Examine the No. 6 screen for any retained cleaning compound and weigh the portion passing the No. 100 sieve and calculate to percentage. If a sieving machine is not available, use the hand sieving procedure of ASTM method D502 or method 2101 of Fed. Test Method Std. No. 536.

4.4.13 Rinsing. Test panels of 2024-T3 aluminum shall be cut 2-1/2 by 2-1/2 inches in size from approximately 0.025-inch-thick sheet conforming to QQ-A-250/4. They shall be polished and cleaned as specified in 4.4.7.1 and 4.4.7.2. Duplicate panels shall be tested. Prepare a 1.0 percent solution of the cleaning compound by the procedure of 4.4.7.3 using 10 grams of cleaning compound. Immerse a panel for 5 minutes in 400 ml of the boiling solution contained in a 600-ml beaker. Remove the panel, suspend at an angle of 45

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degrees until dry, then suspend the panel in about 1 liter of distilled water at 70 ± 2°C for 3 minutes, withdrawing the panel slowly once each minute. Repeat the last step with a fresh sample of distilled water at 70°C. Remove the panel, and let dry at an angle of 45 degrees as before. When dry, examine the panel for residue. Then place a drop of absolute ethyl alcohol on the panel, allow to evaporate, and re-examine for white residue.

4.4.14 Caking in storage. The steam cleaning compound, 125 pounds net weight in a steel drum conforming to type III, 20-gage, of PPP-D-705, fitted with the specified lug cover, shall be stored in an unheated warehouse at Aberdeen Proving Ground, Maryland, for twelve months. The drum shall then be opened and the contents examined for free flowing and granular characteristics. If there is visual evidence of caking or lumping, transfer a single lump of the caked material obtained from the bottom third of the drum, and having a weight between 90 and 110 grams, to a No. 6, U. S. Standard wire cloth sieve conforming to ASTM E11. The sieve shall be placed in a Ro-Tap sieve shaking apparatus, and screened for 5 minutes. Examine the screen for 100 percent passage of material. Lumping of material that is broken up by the sieving operation to permit 100 percent passage through the No. 6 screen will not be cause for failure of this test.

4.4.15 Condition as-received. An as-received container of the steam cleaning compound shall be emptied, and the contents examined for lumping or agglomerating of material. Visual evidence of either shall be cause for rejection of the compound.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A, B, or C, as specified (see 6.2).

5.1.1 Level A. The product shall be packaged in 25-, 125-, and 400-pound quantities. The 25- and 125-pound quantities shall be packaged in drums conforming to PPP-P-704, type II; and PPP-D-705, type III, respectively. The 400-pound quantities shall be packaged in a drum conforming to PPP-D-729, type III. Closure shall be as specified in the applicable container specification.

5.1.2 Level B. The product shall be packaged in 25-, 125-, or 400-pound quantities in fiber drums conforming to PPP-D-723, type I or II, grade A. Drum liners (bags) shall be provided and shall be fabricated from 3-mil thick polyethylene conforming to L-P-378. All bag seams shall be heat sealed. The seals shall meet the heat seal requirements of MIL-P-116. Bag closure may be accomplished by a method other than heat seal when approved by the procuring activity.

5.1.2.1 Alternate level B. The 400-pound quantity shall be packaged in a 55-gallon reconditioned metal drum conforming to PPP-D-732.

5.1.3 Level C. Cleaning compound in quantities as specified (see 6.2) shall be packaged to afford adequate protection against damage during shipment from the supply source to the first receiving activity.

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5.2 Packing. No overpacking is required.

5.3 Marking.

5.3.1 Military agencies. Containers shall be marked in accordance with the requirements of MIL-STD-129.

5.3.2 Civil agencies. Containers shall be marked in accordance with Fed. Std. No. 123.

5.3.3 Directions for use. Unless otherwise specified, each container shall be durably and legibly marked with the directions for use of the compound as follows:

Directions for use. The compound shall be thoroughly dissolved to give a concentration of approximately 10 ounces of the compound per gallon of water, using the soap tank of the cleaning machine, or a separate drum outside the cleaning machine and then transferring to the soap tank of the cleaning machine. The dry compound must be added to hot water in preparing the solution, and not vice versa. Adjust the concentrate solution metering device to obtain the desired results. A nozzle concentration of 0.3 percent (0.4 ounce per gallon) is suitable for average cleaning. When the machine is shut down at the end of a shift, a "blow-down" is essential for retarding scale formation in the coils of the machine.

5.3.4 In addition to the markings specified in 5.3.1, 5.3.2 and 5.3.3, any special marking shall be as required by the order or contract.

6. NOTES

6.1 Intended use. The steam cleaning compound covered by this specification is intended for use in high pressure steam cleaning machines of the continuous tubular coil vapor-generating type, for cleaning of ferrous and nonferrous surfaces of equipment.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Unit quantity required - this compound should be purchased on the basis of net weight in pounds.
(c) Level of packaging required (see 5.1).
(d) Special marking (see 5.3).

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6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are at the time set for opening of bids, qualified for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is the U. S. Army Coating and Chemical Laboratory, Aberdeen Proving Ground, Maryland 21005, and information pertaining to qualification of products may be obtained from that activity.

6.4 Packaging. It has been found that compound packaged in drums permeable to moisture and carbon dioxide tends to cake in storage. Fiber drums especially are susceptible to this action.

6.4.1 Mixing. It has been found that compound prepared by charging the granular ingredients into the blender followed by a short premix and then admixing the liquid surfactant has less tendency to lump or agglomerate than when produced by other orders of mixing.

MILITARY INTEREST:

Custodians:

Army - MR
Navy - SH
Air Force - 68

Review activities:

Army - MI, MR, MU
Navy - AS, SH
Air Force - 68

User activities:

Navy - YD, MC

Preparing activity:

Army - MR

CIVIL AGENCY INTEREST:

GSA

Orders for this publication are to be placed with (
, acting as an agent for the Superintendent of Documents.
'See section 2 of this specification to obtain extra copies and other documents referenced herein. F

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SPECIFICATION ANALYSIS SHEET

Form Approved Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION
P-C-437a, Cleaning Compound, High Pressure (Steam) Cleaner

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD, FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

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Army Materials and Mechanics Research Agency
Watertown, Massachusetts 02172

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Watertown, Massachusetts 02172

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